

Question_3:

Code:

```
1  /*
2  Question No: 3
3  Given an n x n forest where '*' represents a tree and '.'
4  represents an empty square, answer q queries.
5  For each rectangle, count the number of trees inside it.
6  We use a 2D prefix sum so every query can be answered quickly.
7  */
8
9  #include <iostream>
10 using namespace std;
11
12 int a[1005][1005];
13 int main() {
14     int n,q;
15     cin>>n>>q;
16
17     for(int i=1;i<=n;i++) {
18         for(int j=1;j<=n;j++) {
19             char c;
20             cin>>c;
21             a[i][j]=(c=='*')
22                 +a[i-1][j]
23                 +a[i][j-1]
24                 -a[i-1][j-1];
25         }
26     }
27     while(q--) {
28         int x1,y1,x2,y2;
29         cin>>x1>>y1>>x2>>y2;
30
31         cout<<a[x2][y2]
32             -a[x1-1][y2]
33             -a[x2][y1-1]
34             +a[x1-1][y1-1]<<"\n";
35     }
36 }
```

Output:

Test Case_1:

```
4 3
.*..
*.*
**..
****
2 2 3 4
3
3 1 3 1
1
1 1 2 2
2
PS C:\Users\Prabin Yadav\I
```

Test Case_2:

```
4 3
.*.*
*.*
**.*
***.
1 2 3 4
6
3 2 3 4
2
1 4 2 3
0
PS C:\Users\Prabin Yadav\I
```